

# Jorge Becerra Garrido

Curriculum Vitae<sup>1</sup>

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## RESEARCH INTERESTS

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I am mostly interested in quantum topology. The following is a non-exhaustive list of my interests:

- Quantum knot invariants, especially those which arise from ribbon categories/algebras.
- Tangle categories.
- Finite type invariants and the Kontsevich invariant.
- Modular categories, modular functors, modular operads.
- Factorisation homology, skein categories.

## EMPLOYMENT

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**Université Bourgogne Europe** (France). 2024 – present  
Postdoc in the research group of L. Woike.

## EDUCATION

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**Ph.D. in Mathematics**, *University of Groningen* (The Netherlands). 2019 – 2024  
Thesis: *Universal quantum knots invariants*, supervised by R. van der Veen.

**M.Sc. Mathematical Sciences**, *Utrecht University* (The Netherlands). 2017 – 2019  
Honours programme UGC, Cum Laude.  
Thesis: *K-theory with Reality*, supervised by L. Meier.

**B.Sc. Mathematics**, *University of Extremadura* (Spain). 2013 – 2017  
Thesis: *Theory of Relativity*, supervised by J.A. Navarro.

## PUBLICATIONS & PREPRINTS

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1. *Knot invariants from XC-structures on the Sweedler algebra are trivial*. Preprint.
2. *Pivotal Module Categories, Factorization Homology and Modular Invariant Modified Traces* (with Lukas Woike). Preprint.
3. *XC-tangles and universal invariants*. Preprint.
4. *Minimal generating sets of rotational Reidemeister moves* (with Kevin van Helden). Preprint.
5. *A refined functorial universal tangle invariant*. Preprint.
6. *Strictification and non-strictification of monoidal categories*. Preprint.
7. *On Bar-Natan - van der Veen's perturbed Gaussians*, Rev. Real Acad. Cienc. Exactas Fis. Nat. Ser. A-Mat., 118(2):46, 2024.
8. *A combinatorial PROP for bialgebras*. European J. Combin. 124 (2025), Paper No. 104086.
9. *An Introduction to Knot Homology Theories*, TEMat monográficos, 2 (2021): Proceedings of the 3rd BYMAT Conference, pp. 11-14. ISSN: 2660-6003.

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<sup>1</sup>Last updated: 13 February 2026

10. *Finite, Infinite and Twisted Ascent and Descent* (with Jesús MF Castillo), La Gaceta de la Real Sociedad Matemática Española, Vol. 21 (2018), Núm. 3, Págs 497–516.

## RESEARCH TALKS

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- *XC-algebras and quantum knot invariants*, [Congreso Bienal de la Real Sociedad Matemática Española](#), Alicante. Jan 2026
- *Lee homology and the Lee spectral sequence*, [UIC Quantum Topology Seminar](#), Chicago (online). Oct 2025
- *Hopf-algebraic quantum knot invariants*, [GADT groupe de travail](#), Dijon. Nov 2024
- *Universal quantum knot invariants*, [Groningen Quantum Topology Dag](#). May 2024
- *Hopf algebras and knot invariants*, [ICMAT Geometry Seminar](#), Madrid. Jan 2024
- *Drinfeld doubles and Gaussian calculus for universal knot invariants*, [MAPHYAG Seminar](#), Madrid. Jan 2024
- *Quantum invariants of knots and tangles*, [IMUEX Seminar](#), Badajoz. Jan 2024
- *Hopf algebras and the 2-loop polynomial of knots*, [Algebra Seminar](#), Groningen. Dec 2022
- *Three faces of the 2-loop polynomial of knots*, [4th BYMAT](#), online. Nov 2022
- *Universal quantum knot invariants*, [GQT School & Colloquium](#), Zeist. Aug 2022
- *Quantum invariants via perturbed Gaussian expressions*, [Winter Braids XI](#), Dijon Dec 2021
- *A categorical approach to the universal tangle invariant*, [DIAMANT symposium](#), Utrecht. Nov 2021
- *An introduction to knot homology theories*, [3rd BYMAT](#), online. Dec 2020
- *Foundations of stable homotopy theory*, [IMUEX Seminar](#), Badajoz. May 2019
- *Functional inequalities and convergence in diffusion processes*, [13th Meeting on Functional Analysis and Applications](#). Cáceres. Mar 2017

For a list of non-research talks, see [sites.google.com/view/becerra/talks](https://sites.google.com/view/becerra/talks).

## RESEARCH VISITS

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- Universidad Complutense de Madrid, invited by Ángel González Prieto (1 week). Jan 2024

## SELECTED ATTENDANCE

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- *Congreso Bienal de la RSME*, Alicante. Jan 2026
- *Higher Structures: Recent developments and applications*, Hamburg. Sep 2025
- *Matemale spring school*, Matemale. May 2025
- *Algebraic, geometric and quantum topology in Picardie II*, Amiens. Mar 2025
- *Winter Braids XIV*, Bordeaux. Feb 2025
- *Skein days*, Cergy-Pontoise. Dec 2024
- *Algebraic, geometric and quantum topology in Picardie*, Amiens. Mar 2024
- *Swiss Knots 2023*, Regensburg. Sep 2023
- *From Subfactors to Quantum Topology - In memory of Vaughan Jones*, Geneva. Jun 2022
- *Quantum Topology and Geometry (in honor of Vladimir Turaev)*, Paris Jun 2022

- [Winter Braids XI](#), Dijon. Dec 2021
- [Georgia Tech Topology Summer School on 4-manifolds](#), Online. Jul 2021
- [Monoidal and 2-categories in representation theory and categorification](#), Online Dec 2020
- [QUAntum groups, Categorification, Knot invariants, and Soergel bimodules](#), Online. Aug 2020

For a complete list, see [sites.google.com/view/becerra/attendance](https://sites.google.com/view/becerra/attendance).

## TEACHING

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- **Université Bourgogne Europe:**

Teaching Assistant for [Mathématiques pour l'électr. et l'inform. 2](#) (Bachelor). Spring 2026

- **University of Groningen:**

Co-lecturer of [Topics in Topology](#) (Master, universal knot invariants). Spring 2023

Co-lecturer of [Topics in Topology](#) (Master, Kirby calculus). Spring 2022

Co-lecturer of [Topics in Topology](#) (Master, grid homology). Spring 2021

Teaching Assistant for [Multivariable Analysis](#) (Bachelor). Autumn 2020

Co-organiser of the [Knot Theory Seminar](#) (Bach./Master). Spring 2020

Co-organiser of the [Groningen Topology Seminar](#) (Bach./Master). Autumn 2019

- **Utrecht University:**

Teaching Assistant for [Topologie en Meetkunde](#) (Bachelor,  $\pi_1$ ). Spring 2019

Teaching Assistant for [Algebraic Topology](#) (Master,  $H_n$ 's and  $\pi_n$ 's). Autumn 2018

Teaching Assistant for [Topologie en Meetkunde](#) (Bachelor,  $\pi_1$ ). Spring 2018

## BACHELOR'S THESES SUPERVISION

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- Daniël Apol, [Representability of cohomology theories on CW-complexes](#), Groningen. 2021

## SKILLS

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- **Languages:** Spanish (native), English (fluent), French (intermediate), Dutch (basic).
- **Programming languages:**  
Python (with [verified certificate](#) by HarvardX), including SAGE, Numpy, Pandas, Matplotlib, NetworkX and Scikit-learn.  
Mathematica, C, R-Commander, Fortran, L<sup>A</sup>T<sub>E</sub>X, Linux.
- **Teaching:** Teacher training certificate.

## ORGANISATION

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- Co-organiser of the talk series [Outils algébriques en théorie conforme des champs](#), Dijon 2025 - pres
- Co-organiser of the [Groupe de Travail sur TQFTs \(non-\)semisimples](#), Dijon. 2025 - pres
- Co-organiser of the workshop [Groningen Quantum Topology Dag](#) May 2024
- Co-organiser of the [Quantum Mechanics bookclub](#), Groningen. 2022
- Co-organiser of the [Knot homology theories reading group](#), Groningen. 2020
- Co-organiser of the [PhD Seminar](#), Groningen. 2020 - 2021

## GRANTS

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- [Research Assistant](#), Department of Mathematics, University of Extremadura (Badajoz, Spain). €2,000. 2016–2017

- Fee waiver for talented students, *Summer School on Geometry*, Utrecht (The Netherlands). €400 2016
- Twelve university course tuition fee waivers for top student of a course. €1,200 2013–2016
- One-year university tuition fee waiver for top high-school student. €1,000 2013

#### PROFESSIONAL ACTIVITIES

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- Speaker at various outreach activities, e.g. the *European Researchers' Night* at the University of Extremadura. 2014 - 2017
- Private tutor of English, Mathematics and Physics for high school students 2013 – 2017